

CYBERSECURITY ATTACK PATH VISUALIZATION SYSTEM

AI-Based Graph Search Algorithms Simulator

“Modeling network security as intelligent graph traversal for attack detection and path optimization.”

PROJECT REPORT (LAB MID SUBMISSION)

Academic Information

- **Course Title:** *Artificial Intelligence*
- **Course Code:** *CSC 262*
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- **Institution:** *CUI ATD*

Supervision

Submitted To:

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Submission Details

- **Submission Date:** 30 May 2026
- **Session:** Spring 2026
- **Type:** Lab Mid Project

Project Abstract

This project presents a **graph-based cybersecurity simulation system** that models a computer network as an interconnected graph of nodes (systems) and edges (connections).

It applies **AI search algorithms** to analyze possible attack paths an adversary may take to reach sensitive systems.

The system visually demonstrates:

- Breadth First Search (BFS)
- Depth First Search (DFS)
- A* Algorithm
- UCS
- Hill Climbing
- Node traversal behavior
- Attack propagation in a network

The goal is to improve understanding of **pathfinding algorithms in cybersecurity threat analysis**.

Core Concepts Implemented

- ✓ Graph Theory Modeling
- ✓ AI Search Algorithms

- ✓ Attack Path Simulation
- ✓ Step-by-step Node Traversal
- ✓ Real-time Visualization (Canvas-based)

Technologies Used

- HTML5 (Structure)
- CSS3 (UI Styling)
- JavaScript (Algorithm Logic)
- Canvas API (Graph Rendering)
- Network Modeling Concepts
- Google Colab

Project Objectives

- To simulate real-world cyber network structures
- To apply BFS & DFS in security analysis
- To visualize algorithm behavior dynamically
- To understand attack path discovery techniques

Expected Output

- Interactive graph visualization
- Step-by-step traversal animation
- Highlighted nodes during execution
- Clear attack path representation

System Insight

This project bridges Artificial Intelligence and Cybersecurity by demonstrating how search algorithms can be used to analyze vulnerabilities in network systems.

Live System Access

BFS, DFS, A* and UCS Implementation (Graph-Based AI Simulation)

 **Launch Project** → [Cybersecurity AI Graph Simulator](#)

MinMax and Alpha Beta Pruning

 **Launch Project** → [Cybersecurity AI Graph Simulator](#)

⌵ Libraries

```
!pip install networkx matplotlib pandas
```

```
Requirement already satisfied: networkx in /usr/local/lib/python3.12/dist-packages (3.6.1)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matpl
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotli
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Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplo
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib)
```

```
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matpl
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from m
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateut
```

```
%matplotlib inline
```

```
import networkx as nx
import matplotlib.pyplot as plt
from collections import deque
import heapq
import time
import pandas as pd
```

✓ Cybersecurity Attack Network Graph

```
import networkx as nx
import matplotlib.pyplot as plt

# -----
# Create Cyber Security Network
# -----
G = nx.Graph()

edges = [
    ("Attacker", "Firewall", 2),
    ("Attacker", "MailServer", 4),
    ("Firewall", "WebServer", 3),
    ("Firewall", "AdminPC", 5),
    ("MailServer", "WebServer", 2),
    ("MailServer", "BackupServer", 6),
    ("WebServer", "Database", 1),
    ("WebServer", "CloudNode", 4),
    ("AdminPC", "Database", 2),
    ("BackupServer", "CloudNode", 2),
    ("CloudNode", "Database", 3)
]

for u, v, w in edges:
    G.add_edge(u, v, weight=w)
start = "Attacker"
goal = "Database"

# -----
# Layout (professional look)
# -----
pos = nx.spring_layout(G, seed=42)

# -----
# Node Styling
# -----
node_colors = []
for node in G.nodes():
    if node == "Attacker":
        node_colors.append("red")
    elif node == "Database":
        node_colors.append("green")
    else:
        node_colors.append("skyblue")

# -----
# Draw Graph
# -----
plt.figure(figsize=(12, 8))
```

```

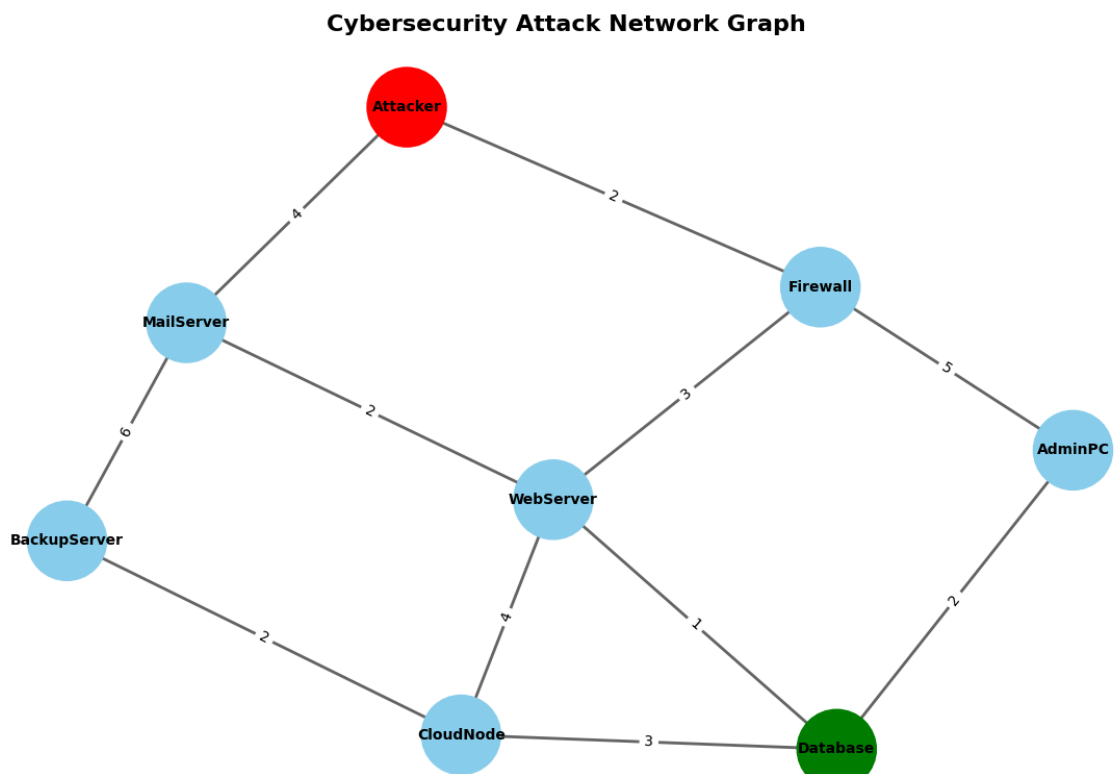
nx.draw_networkx_nodes(G, pos, node_color=node_colors, node_size=3000)
nx.draw_networkx_edges(G, pos, width=2, alpha=0.6)
nx.draw_networkx_labels(G, pos, font_size=10, font_weight="bold")

edge_labels = nx.get_edge_attributes(G, "weight")
nx.draw_networkx_edge_labels(G, pos, edge_labels=edge_labels)

# -----
# Title
# -----
plt.title("Cybersecurity Attack Network Graph ", fontsize=16, fontweight="bold")

plt.axis("off")
plt.tight_layout()
plt.show()

```



✓ BFS Implementation

```

import networkx as nx
import matplotlib.pyplot as plt
from collections import deque

# -----
# Create Cyber Security Network
# -----
G = nx.Graph()

```

```

edges = [
    ("Attacker", "Firewall", 2),
    ("Attacker", "MailServer", 4),
    ("Firewall", "WebServer", 3),
    ("Firewall", "AdminPC", 5),
    ("MailServer", "WebServer", 2),
    ("MailServer", "BackupServer", 6),
    ("WebServer", "Database", 1),
    ("WebServer", "CloudNode", 4),
    ("AdminPC", "Database", 2),
    ("BackupServer", "CloudNode", 2),
    ("CloudNode", "Database", 3)
]

for u, v, w in edges:
    G.add_edge(u, v, weight=w)

# -----
# BFS Implementation
# -----
start_node = "Attacker"

visited = set()
level = {start_node: 0}
parent = {}

queue = deque([start_node])
visited.add(start_node)

while queue:
    node = queue.popleft()

    for neighbor in G.neighbors(node):
        if neighbor not in visited:
            visited.add(neighbor)
            queue.append(neighbor)
            level[neighbor] = level[node] + 1
            parent[neighbor] = node

# -----
# BFS Tree Edges
# -----
bfs_edges = [(parent[n], n) for n in parent]

# -----
# Layout
# -----
pos = nx.spring_layout(G, seed=42)

# -----
# Node Colors by BFS Level
# -----
color_map = []
for node in G.nodes():
    if node == start_node:
        color_map.append("red")          # Attacker
    elif node not in level:
        color_map.append("gray")
    else:
        # gradient effect by level
        if level[node] == 1:
            color_map.append("orange")
        elif level[node] == 2:
            color_map.append("skyblue")
        else:
            color_map.append("lightgreen")

# -----
# Draw Graph

```

```
# -----
plt.figure(figsize=(12, 8))

# All edges (background network)
nx.draw_networkx_edges(G, pos, alpha=0.3, width=2)

# BFS tree edges (highlighted)
nx.draw_networkx_edges(G, pos, edgelist=bfs_edges, edge_color="red", width=3)

# Nodes
nx.draw_networkx_nodes(G, pos, node_color=color_map, node_size=3000)

# Labels
nx.draw_networkx_labels(G, pos, font_size=10, font_weight="bold")

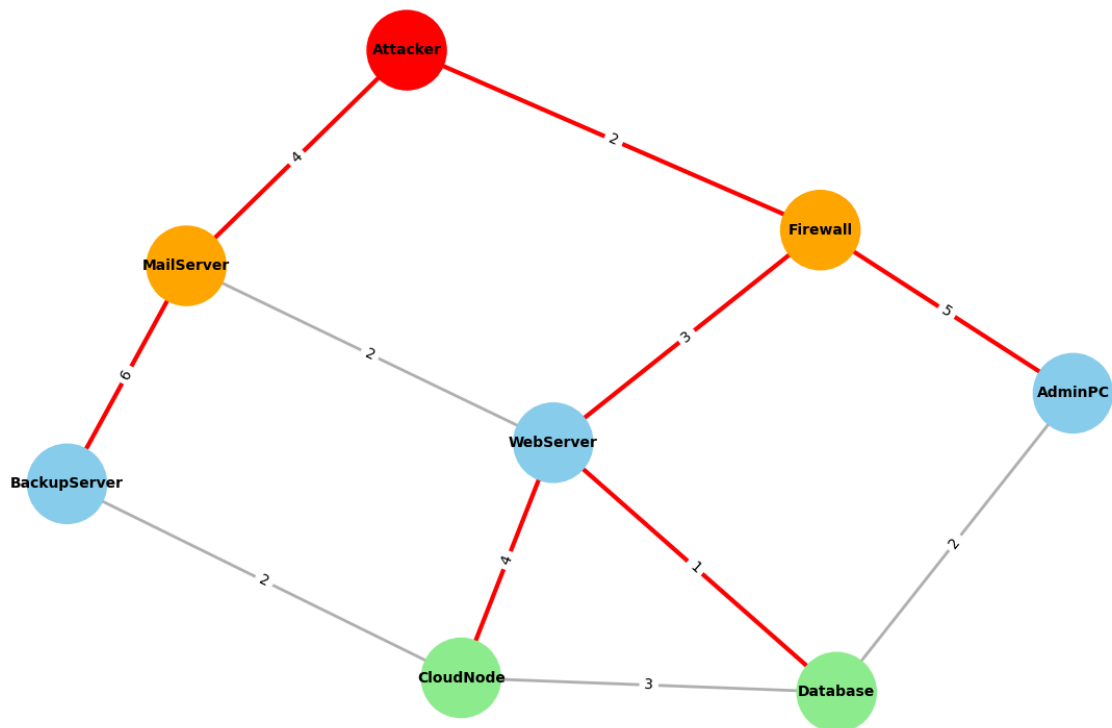
# Edge weights
edge_labels = nx.get_edge_attributes(G, "weight")
nx.draw_networkx_edge_labels(G, pos, edge_labels=edge_labels)

# -----
# Title
# -----
plt.title("Cybersecurity BFS Attack Propagation Tree", fontsize=16, fontweight="bold")

plt.axis("off")
plt.tight_layout()
plt.show()

# -----
# Print BFS Order
# -----
print("BFS Traversal Order from Attacker:")
print(" -> ".join(list(level.keys())))
```

Cybersecurity BFS Attack Propagation Tree



BFS Traversal Order from Attacker:

Attacker -> Firewall -> MailServer -> WebServer -> AdminPC -> BackupServer -> Database -> CloudNode

DFS Implementation

```

import networkx as nx
import matplotlib.pyplot as plt
import time
# Create Cyber Security Network
G = nx.Graph()

edges = [
    ("Attacker", "Firewall", 2),
    ("Attacker", "MailServer", 4),
    ("Firewall", "WebServer", 3),
    ("Firewall", "AdminPC", 5),
    ("MailServer", "WebServer", 2),
    ("MailServer", "BackupServer", 6),
    ("WebServer", "Database", 1),
    ("WebServer", "CloudNode", 4),
    ("AdminPC", "Database", 2),
    ("BackupServer", "CloudNode", 2),
    ("CloudNode", "Database", 3)
]

for u, v, w in edges:
    G.add_edge(u, v, weight=w)
# DFS Implementation (Deterministic)

```

```

start_node = "Attacker"
goal_node = "Database"

start_time = time.time()

stack = [(start_node, [start_node])]
visited = set()

nodes_expanded = 0
final_path = None

while stack:
    node, path = stack.pop()

    if node in visited:
        continue

    visited.add(node)
    nodes_expanded += 1
    # Goal check
    if node == goal_node:
        final_path = path
        break
    neighbors = sorted(G.neighbors(node), reverse=True)

    for neighbor in neighbors:
        if neighbor not in visited:
            stack.append((neighbor, path + [neighbor]))

end_time = time.time()
execution_time_ms = (end_time - start_time) * 1000

total_cost = 0
if final_path:
    for i in range(len(final_path) - 1):
        u, v = final_path[i], final_path[i+1]
        total_cost += G[u][v]['weight']
print("===== DFS RESULT =====")
print("Discovered Path:", " → ".join(final_path) if final_path else "No Path Found")
print("Total Cost:", total_cost)
print("Nodes Expanded:", nodes_expanded)
print("Execution Time (ms):", round(execution_time_ms, 4))

pos = nx.spring_layout(G, seed=42)

plt.figure(figsize=(12, 8))

# Node colors
node_colors = []
for node in G.nodes():
    if node == "Attacker":
        node_colors.append("red")
    elif node == "Database":
        node_colors.append("green")
    else:
        node_colors.append("skyblue")

# Draw all nodes and edges
nx.draw_networkx_nodes(G, pos, node_color=node_colors, node_size=3000)
nx.draw_networkx_labels(G, pos, font_size=10, font_weight="bold")
nx.draw_networkx_edges(G, pos, width=1.5, alpha=0.4)

# Highlight DFS path
if final_path:
    path_edges = list(zip(final_path, final_path[1:]))
    nx.draw_networkx_edges(G, pos, edgelist=path_edges, width=4, edge_color="red")

# Edge labels (weights)
edge_labels = nx.get_edge_attributes(G, "weight")
nx.draw_networkx_edge_labels(G, pos, edge_labels=edge_labels)

```



```
plt.title("DFS Attack Path Visualization (Cybersecurity Graph)", fontsize=14, fontweight="bold")
plt.axis("off")
plt.show()
```

===== DFS RESULT =====

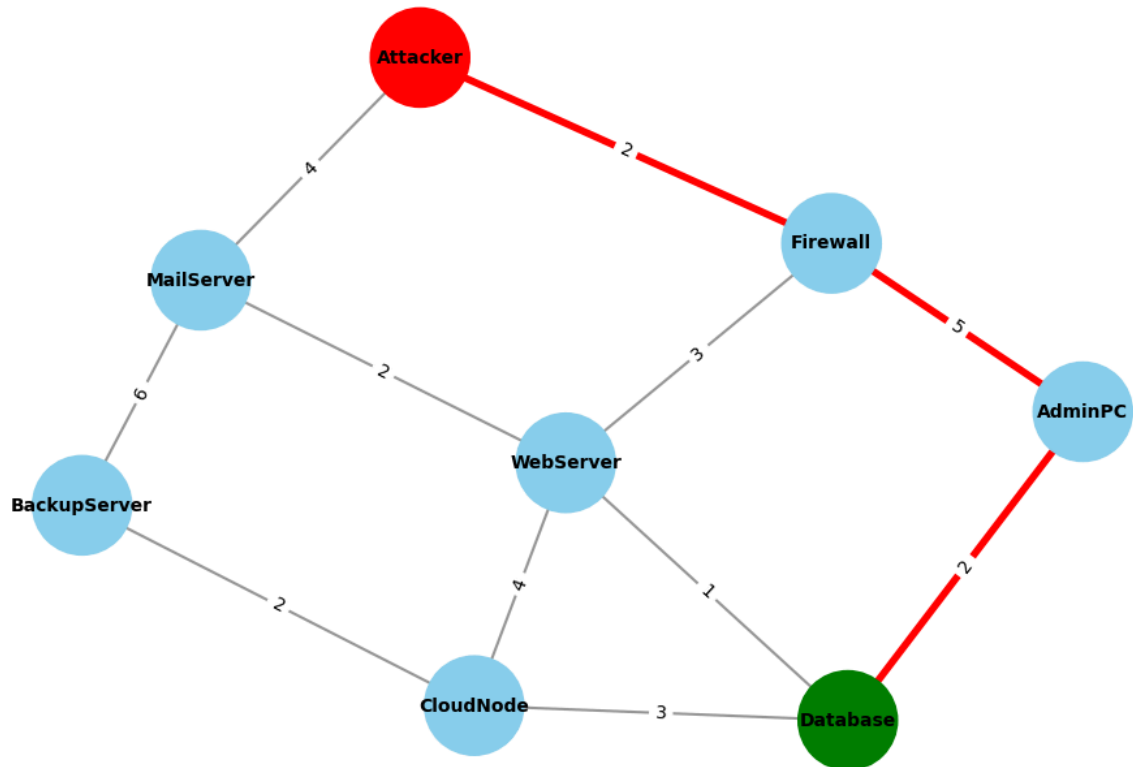
Discovered Path: Attacker → Firewall → AdminPC → Database

Total Cost: 9

Nodes Expanded: 4

Execution Time (ms): 0.4981

DFS Attack Path Visualization (Cybersecurity Graph)



✓ A* Algorithm

```
import networkx as nx
import matplotlib.pyplot as plt
# CREATE GRAPH
G = nx.Graph()

edges = [

    ("Attacker", "Firewall", 2),
    ("Attacker", "MailServer", 4),

    ("Firewall", "WebServer", 3),
    ("Firewall", "AdminPC", 5),

    ("MailServer", "WebServer", 2),
    ("MailServer", "BackupServer", 6),

    ("BackupServer", "CloudNode", 2),

    ("WebServer", "CloudNode", 4),
```

```

        ("WebServer", "Database", 1),

        ("CloudNode", "Database", 3),

        ("AdminPC", "Database", 2)
    ]

    for u, v, w in edges:
        G.add_edge(u, v, weight=w)
    # DEFINE START & GOAL
    start = "Attacker"
    goal = "Database"

    astar_path = [
        "Attacker",
        "Firewall",
        "WebServer",
        "Database"
    ]

    # -----
    # GRAPH POSITIONING
    # -----

    pos = nx.spring_layout(G, seed=42)

    # -----
    # NODE COLORS
    # -----

    node_colors = []

    for node in G.nodes():

        if node == start:
            node_colors.append("red")

        elif node == goal:
            node_colors.append("green")

        elif node in astar_path:
            node_colors.append("orange")

        else:
            node_colors.append("skyblue")
    # DRAW GRAPH
    plt.figure(figsize=(12,8))

    nx.draw(
        G,
        pos,
        with_labels=True,
        node_color=node_colors,
        node_size=2500,
        font_weight="bold",
        edge_color="gray"
    )

    # Edge labels
    labels = nx.get_edge_attributes(G, "weight")

    nx.draw_networkx_edge_labels(
        G,
        pos,
        edge_labels=labels
    )

    # Highlight A* path
    edges_path = list(zip(astar_path, astar_path[1:]))

```

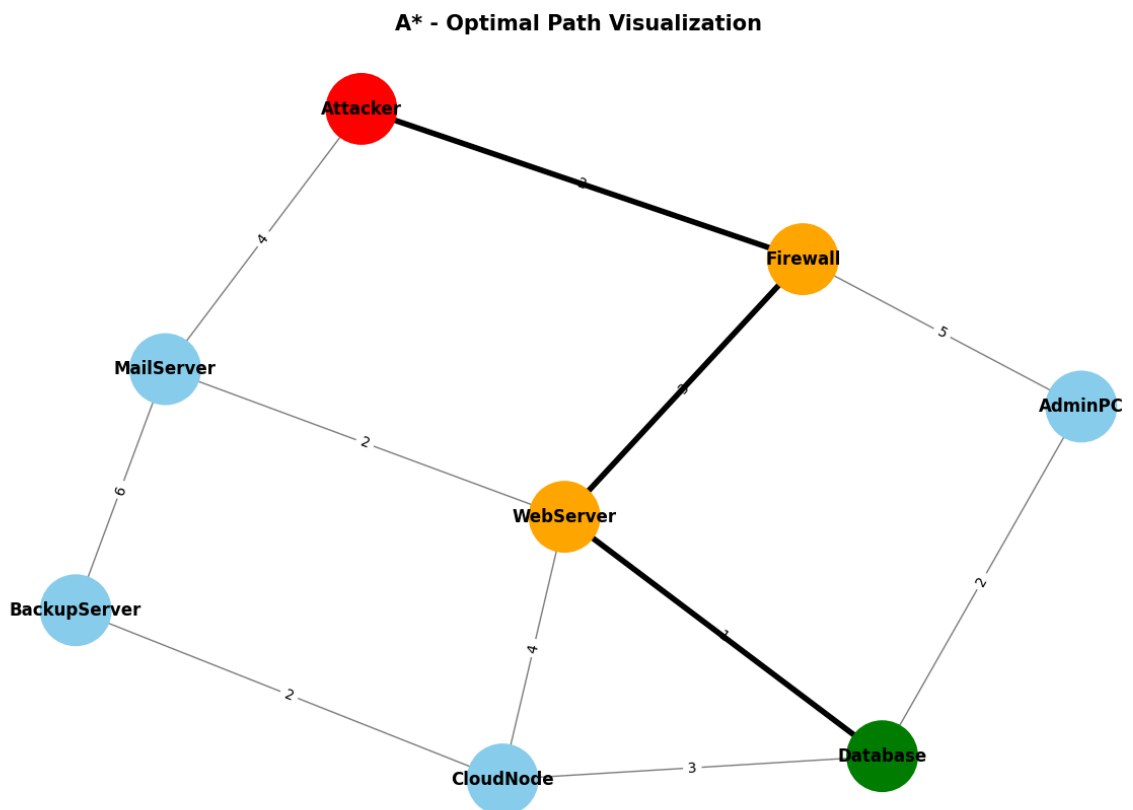
```

nx.draw_networkx_edges(
    G,
    pos,
    edgelist=edges_path,
    width=4,
    edge_color="black"
)

plt.title(
    "A* - Optimal Path Visualization",
    fontsize=15,
    fontweight="bold"
)

plt.axis("off")
plt.show()

```



✓ UCS Implementation

```

def ucs():
    start_time = time.time()

    pq = [(0, start)]
    visited = set()
    parent = {}
    cost = {start: 0}

```

```

nodes_expanded = 0

while pq:
    g, node = heapq.heappop(pq)

    if node in visited:
        continue

    visited.add(node)
    nodes_expanded += 1

    if node == goal:
        break

    for n in G.neighbors(node):
        new_cost = g + G[node][n]['weight']

        if n not in cost or new_cost < cost[n]:
            cost[n] = new_cost
            parent[n] = node
            heapq.heappush(pq, (new_cost, n))

path = []
cur = goal
while cur != start:
    path.append(cur)
    cur = parent[cur]
path.append(start)
path.reverse()

return path, cost[goal], nodes_expanded, (time.time()-start_time)*1000

ucs_path, ucs_cost, ucs_nodes, ucs_time = ucs()

```

```

ucs_path, ucs_cost, ucs_nodes, ucs_time = ucs()

print("\n===== UCS RESULTS =====\n")

print(" Discovered Path:")
print(" -> ".join(ucs_path))
print("\n Total Cost:", ucs_cost)
print(" Nodes Expanded:", ucs_nodes)
print(" Execution Time (ms):", round(ucs_time, 4))

```

===== UCS RESULTS =====

Discovered Path:
Attacker -> Firewall -> WebServer -> Database

Total Cost: 6
Nodes Expanded: 5
Execution Time (ms): 0.0451

```

def draw_ucs_path():
    plt.figure(figsize=(12, 8))
    node_colors = []
    for node in G.nodes():
        if node == start:
            node_colors.append("red")      # Start
        elif node == goal:
            node_colors.append("green")    # Goal
        elif node in ucs_path:
            node_colors.append("orange")   # UCS path
        else:
            node_colors.append("skyblue")

    pos = nx.spring_layout(G, seed=42)

```

```

# draw nodes + edges
nx.draw(G, pos,
        with_labels=True,
        node_color=node_colors,
        node_size=3000,
        font_weight="bold")

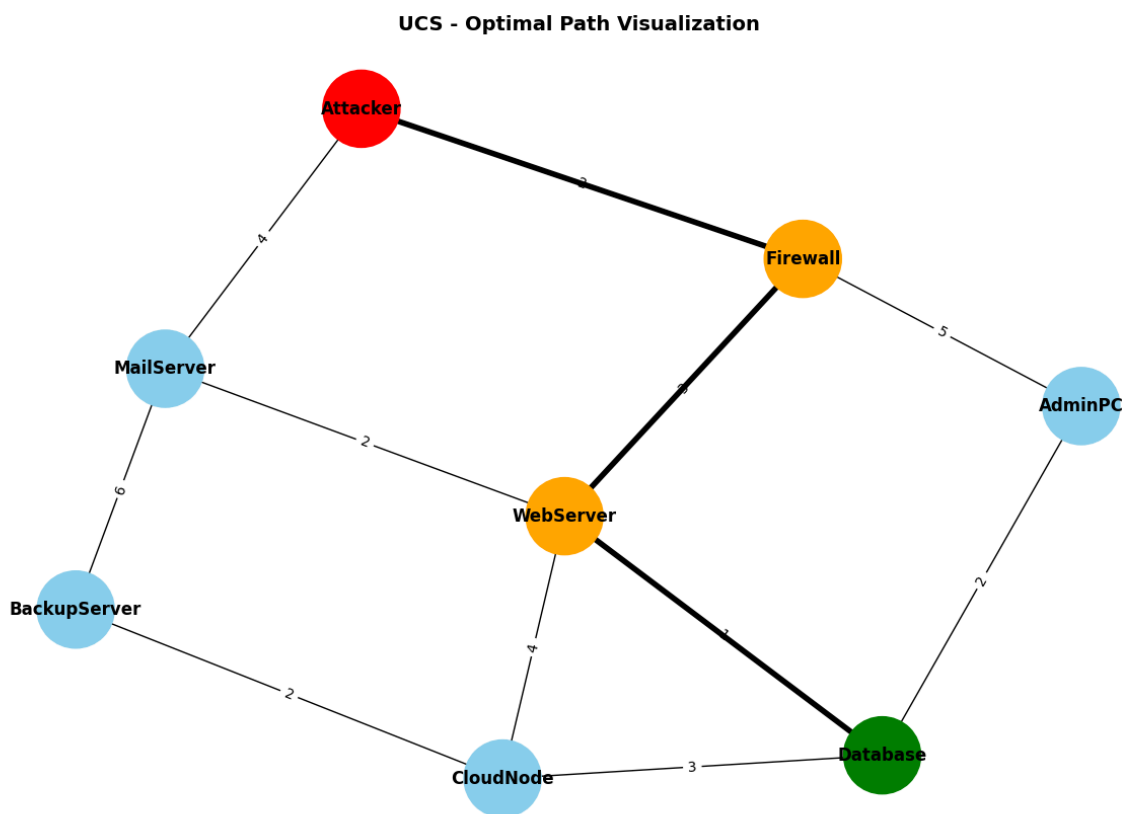
# edge weights
labels = nx.get_edge_attributes(G, "weight")
nx.draw_networkx_edge_labels(G, pos, edge_labels=labels)

# highlight UCS path
path_edges = list(zip(ucs_path, ucs_path[1:]))
nx.draw_networkx_edges(G, pos,
                       edgelist=path_edges,
                       width=4,
                       edge_color="black")

plt.title("UCS - Optimal Path Visualization", fontsize=14, fontweight="bold")
plt.axis("off")
plt.show()

draw_ucs_path()

```



✓ Hill Climbing Algorithm

```

def hill_climbing():
    start_time = time.time()

    current = start
    path = [current]
    visited = set([current])
    nodes_expanded = 0

    while current != goal:
        nodes_expanded += 1

        neighbors = list(G.neighbors(current))

        next_node = None
        best_h = float("inf")

        for n in neighbors:
            if n not in visited and h[n] < best_h:
                best_h = h[n]
                next_node = n

        if next_node is None:
            break

        current = next_node
        visited.add(current)
        path.append(current)

    cost = sum(G[path[i]][path[i+1]]['weight'] for i in range(len(path)-1))

    return path, cost, nodes_expanded, (time.time()-start_time)*1000

hc_path, hc_cost, hc_nodes, hc_time = hill_climbing()
hc_path, hc_cost, hc_nodes, hc_time = hill_climbing()

print("\n===== HILL CLIMBING RESULTS =====\n")
print(" Discovered Path:")
print(" -> ".join(hc_path))
print("\n Total Cost:", hc_cost)
print(" Nodes Expanded:", hc_nodes)
print(" Execution Time (ms):", round(hc_time, 4))

```

```

===== HILL CLIMBING RESULTS =====

Discovered Path:
Attacker -> MailServer -> WebServer -> Database

Total Cost: 7
Nodes Expanded: 3
Execution Time (ms): 0.0169

```

```

def draw_hc_path():
    plt.figure(figsize=(12, 8))

    node_colors = []
    for node in G.nodes():
        if node == start:
            node_colors.append("red")
        elif node == goal:
            node_colors.append("green")
        elif node in hc_path:
            node_colors.append("orange")
        else:
            node_colors.append("skyblue")

    pos = nx.spring_layout(G, seed=42)

    nx.draw(G, pos,

```

```

with_labels=True,
node_color=node_colors,
node_size=3000,
font_weight="bold")

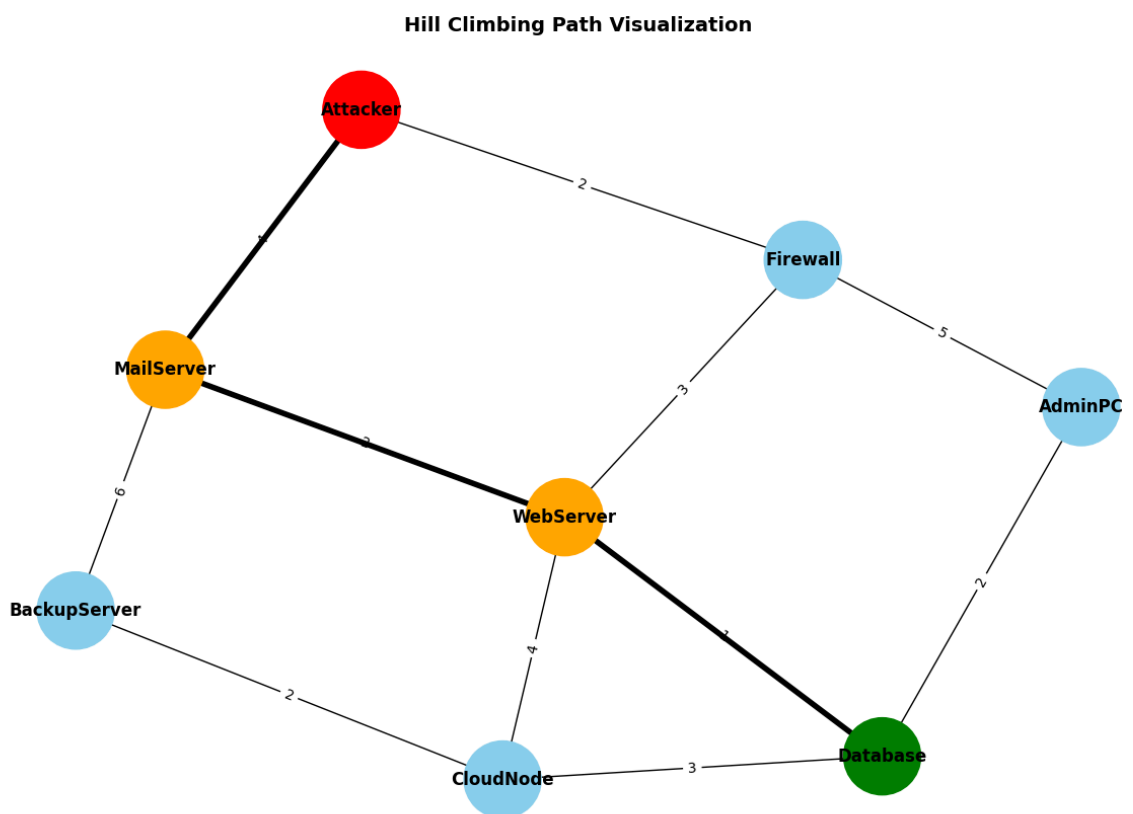
labels = nx.get_edge_attributes(G, "weight")
nx.draw_networkx_edge_labels(G, pos, edge_labels=labels)

path_edges = list(zip(hc_path, hc_path[1:]))
nx.draw_networkx_edges(G, pos,
                      edgelist=path_edges,
                      width=4,
                      edge_color="black")

plt.title("Hill Climbing Path Visualization", fontsize=14, fontweight="bold")
plt.axis("off")
plt.show()

draw_hc_path()

```



✓ Local Maxima

```

import networkx as nx
G = nx.Graph()

```

```

edges = [

    ("Attacker", "Firewall", 2),
    ("Attacker", "MailServer", 4),

    ("Firewall", "WebServer", 3),
    ("Firewall", "AdminPC", 5),

    ("MailServer", "WebServer", 2),
    ("MailServer", "BackupServer", 6),

    ("WebServer", "Database", 4),

    ("AdminPC", "Database", 7),

    ("BackupServer", "Database", 3)
]

# Add edges into graph
for u, v, w in edges:
    G.add_edge(u, v, weight=w)
heuristic = {

    "Attacker": 10,
    "Firewall": 8,

    # Local maxima node
    "MailServer": 2,

    # WORSE neighbors
    "WebServer": 9,
    "BackupServer": 7,

    "AdminPC": 6,

    # Goal
    "Database": 0
}

def hill_climbing(graph, start, goal, heuristic):

    current = start
    path = [current]
    visited = set()
    while current != goal:

        visited.add(current)

        print(f"\n Current Node: {current}")
        print(f"Heuristic Value: {heuristic[current]}")

        neighbors = list(graph.neighbors(current))

        # Remove visited nodes
        neighbors = [n for n in neighbors if n not in visited]

        if not neighbors:

            print("\n FAILURE: No more neighbors available.")
            return path

        print("\n Neighbor Analysis:")

        best_neighbor = None
        best_h = heuristic[current]

        for neighbor in neighbors:

            h_value = heuristic[neighbor]

            print(f"    → {neighbor} : h = {h_value}")

```



```

        if h_value < best_h:

            best_h = h_value
            best_neighbor = neighbor
        if best_neighbor is None:

            print("\n⚠ LOCAL MAXIMA DETECTED ⚠")

            print(f"""
Hill Climbing FAILED because:

• Current node = {current}
• All neighboring nodes have WORSE heuristic values
• No better move exists

Algorithm becomes stuck at local optimum.
Goal node could not be reached.
""")

            return path

        print(f"\n Moving to Better Neighbor: {best_neighbor}")

        current = best_neighbor
        path.append(current)

        print("\n GOAL REACHED SUCCESSFULLY!")

        return path
hill_path = hill_climbing(G, "Attacker", "Database", heuristic)

print("\nFinal Path:", hill_path)

```

```

=====
HILL CLIMBING SEARCH STARTED
=====

```

```

Current Node: Attacker
Heuristic Value: 10

```

```

Neighbor Analysis:
  → Firewall : h = 8
  → MailServer : h = 2

```

```

Moving to Better Neighbor: MailServer

```

```

Current Node: MailServer
Heuristic Value: 2

```

```

Neighbor Analysis:
  → WebServer : h = 9
  → BackupServer : h = 7

```

```

⚠ LOCAL MAXIMA DETECTED ⚠

```

```

Hill Climbing FAILED because:

```

- Current node = MailServer
- All neighboring nodes have WORSE heuristic values
- No better move exists

```

Algorithm becomes stuck at local optimum.
Goal node could not be reached.

```

```

Final Path: ['Attacker', 'MailServer']

```

✓ Graphical Representation

```

import networkx as nx
import matplotlib.pyplot as plt
G = nx.Graph()

edges = [

    ("Attacker", "Firewall", 2),
    ("Attacker", "MailServer", 4),

    ("Firewall", "WebServer", 3),
    ("Firewall", "AdminPC", 5),

    ("MailServer", "WebServer", 2),
    ("MailServer", "BackupServer", 6),

    ("BackupServer", "CloudNode", 2),

    ("WebServer", "CloudNode", 4),
    ("WebServer", "Database", 1),

    ("CloudNode", "Database", 3),

    ("AdminPC", "Database", 2)
]

for u, v, w in edges:
    G.add_edge(u, v, weight=w)
heuristic = {

    "Attacker": 10,
    "Firewall": 8,

    # Local maxima node
    "MailServer": 2,

    # Worse neighbors
    "WebServer": 9,
    "BackupServer": 7,

    "CloudNode": 6,
    "AdminPC": 5,

    "Database": 0
}

def hill_climbing(graph, start, goal, heuristic):

    current = start
    path = [current]
    visited = set()
    while current != goal:

        visited.add(current)

        print(f"\n 📍 Current Node: {current}")
        print(f"Heuristic Value: {heuristic[current]}")

        neighbors = list(graph.neighbors(current))
        neighbors = [n for n in neighbors if n not in visited]

        print(f"\n 🗺 Neighbor Analysis:")

        best_neighbor = None
        best_h = heuristic[current]

        for neighbor in neighbors:

```

```

        h_value = heuristic[neighbor]

        print(f"    → {neighbor} : h = {h_value}")

        if h_value < best_h:

            best_h = h_value
            best_neighbor = neighbor
# LOCAL MAXIMA DETECTION
            if best_neighbor is None:

                print("\n⚠ LOCAL MAXIMA DETECTED ⚠")

            print(f"""
Hill Climbing FAILED because:

• Current node = {current}
• All neighboring nodes have WORSE heuristic values
• No better move exists

Algorithm becomes stuck at local optimum.
Goal node could not be reached.
""")

            return path, current

        print(f"\n✅ Moving to Better Neighbor: {best_neighbor}")

        current = best_neighbor
        path.append(current)

        print(f"\n🎯 GOAL REACHED SUCCESSFULLY!")

        return path, current

# -----
# RUN ALGORITHM
# -----

start = "Attacker"
goal = "Database"

hill_path, failed_node = hill_climbing(
    G,
    start,
    goal,
    heuristic
)

print(f"\nFinal Path:", hill_path)

# -----
# VISUALIZATION
# -----

plt.figure(figsize=(13,8))

pos = nx.spring_layout(G, seed=42)

# -----
# NODE COLORS
# -----

node_colors = []

for node in G.nodes():

    # Start node
    if node == start:

```

```

        node_colors.append("red")

    # Goal node
    elif node == goal:
        node_colors.append("green")

    # Failure node
    elif node == failed_node:
        node_colors.append("purple")

    # Traversed path
    elif node in hill_path:
        node_colors.append("orange")

    else:
        node_colors.append("skyblue")

# -----
# DRAW GRAPH
# -----

nx.draw(
    G,
    pos,
    with_labels=True,
    node_color=node_colors,
    node_size=3000,
    font_weight="bold",
    edge_color="gray"
)

# Edge labels
labels = nx.get_edge_attributes(G, "weight")

nx.draw_networkx_edge_labels(
    G,
    pos,
    edge_labels=labels
)

# Highlight Hill Climbing Path
path_edges = list(zip(hill_path, hill_path[1:]))

nx.draw_networkx_edges(
    G,
    pos,
    edgelist=path_edges,
    width=5,
    edge_color="black"
)

# -----
# TITLE
# -----

plt.title(
    "Hill Climbing Failure Visualization (Local Maxima)",
    fontsize=16,
    fontweight="bold"
)

# Failure message on graph
plt.text(
    0.02,
    0.02,
    "⚠ Hill Climbing Stuck at Local Maxima",
    fontsize=12,
    color="purple",
    transform=plt.gca().transAxes,
    bbox=dict(facecolor='white', alpha=0.8)

```

```
)
```

```
plt.axis("off")
```

```
plt.show()
```

=====

HILL CLIMBING SEARCH STARTED

=====

📍 Current Node: Attacker
Heuristic Value: 10

🔍 Neighbor Analysis:
→ Firewall : h = 8
→ MailServer : h = 2

✅ Moving to Better Neighbor: MailServer

📍 Current Node: MailServer
Heuristic Value: 2

🔍 Neighbor Analysis:
→ WebServer : h = 9
→ BackupServer : h = 7

⚠️ LOCAL MAXIMA DETECTED ⚠️

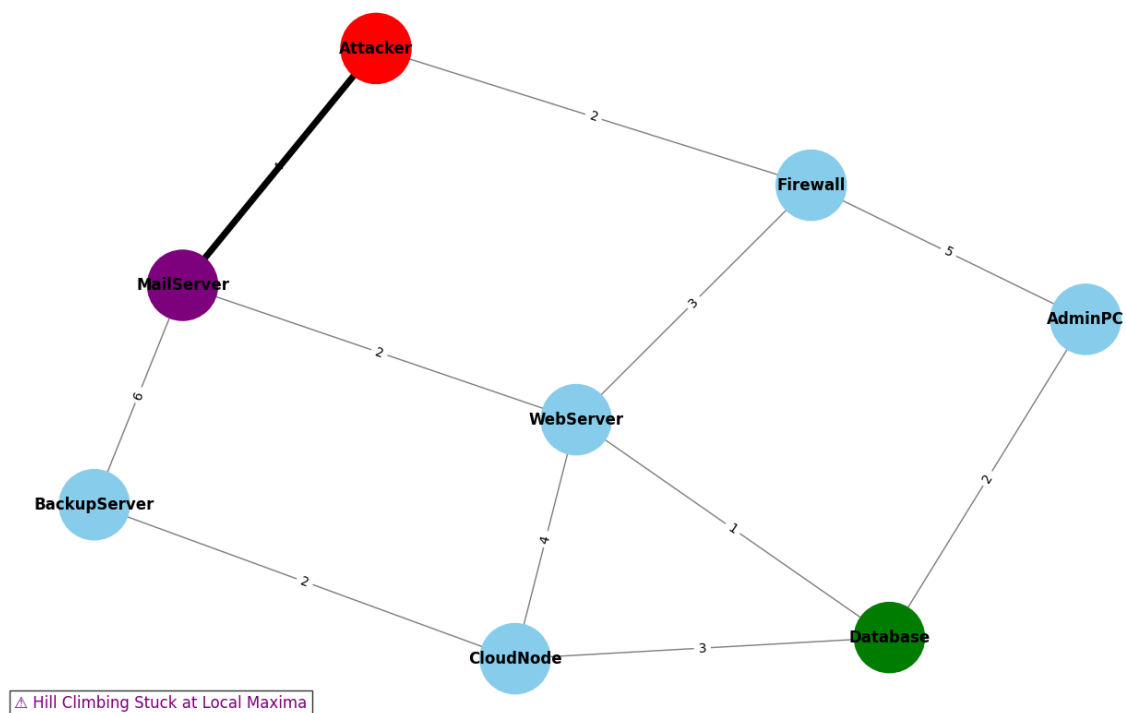
Hill Climbing FAILED because:

- Current node = MailServer
- All neighboring nodes have WORSE heuristic values
- No better move exists

Algorithm becomes stuck at local optimum.
Goal node could not be reached.

Final Path: ['Attacker', 'MailServer']

Hill Climbing Failure Visualization (Local Maxima)



✓ MINIMAX ALGORITHM

```

import networkx as nx
import matplotlib.pyplot as plt

# -----
# GAME TREE GRAPH
# -----

G = nx.DiGraph()

edges = [

    ("Attacker", "Firewall"),
    ("Attacker", "MailServer"),

    ("Firewall", "Node3"),
    ("Firewall", "Node5"),

    ("MailServer", "Node2"),
    ("MailServer", "Node9")
]

G.add_edges_from(edges)

# Terminal values
values = {
    "Node3": 3,
    "Node5": 5,
    "Node2": 2,
    "Node9": 9
}

# -----
# MINIMAX WITH PATH TRACKING
# -----

def minimax(node, maximizingPlayer):

    # Terminal node
    if node in values:
        return values[node], [node]

    children = list(G.successors(node))

    # MAX PLAYER (Attacker)
    if maximizingPlayer:

        best_value = -float("inf")
        best_path = []

        for child in children:

            val, path = minimax(child, False)

            if val > best_value:
                best_value = val
                best_path = [node] + path

        return best_value, best_path

    # MIN PLAYER (Defender)
    else:

        best_value = float("inf")
        best_path = []

```

```

        for child in children:

            val, path = minimax(child, True)

            if val < best_value:
                best_value = val
                best_path = [node] + path

        return best_value, best_path

# -----
# RUN MINIMAX
# -----

result, best_path = minimax("Attacker", True)

print("\n=====")
print(" MINIMAX RESULT ")
print("=====")

print("Optimal Value:", result)
print("Best Path:", " → ".join(best_path))

# -----
# VISUALIZATION
# -----

plt.figure(figsize=(10,7))

pos = {

    "Attacker": (0,4),

    "Firewall": (-2,2),
    "MailServer": (2,2),

    "Node3": (-3,0),
    "Node5": (-1,0),

    "Node2": (1,0),
    "Node9": (3,0)
}

# Node colors
node_colors = []

for node in G.nodes():

    if node == "Attacker":
        node_colors.append("red")

    elif node in ["Firewall", "MailServer"]:
        node_colors.append("orange")

    else:
        node_colors.append("lightgreen")

# Draw full graph
nx.draw(
    G,
    pos,
    with_labels=True,
    node_size=3000,
    node_color=node_colors,
    font_weight="bold",
    arrows=True
)

```



```
# Terminal values on nodes
labels = {node: str(values[node]) for node in values}

nx.draw_networkx_labels(
    G,
    pos,
    labels=labels,
    font_color="blue"
)

# -----
# HIGHLIGHT BEST PATH (BLUE)
# -----

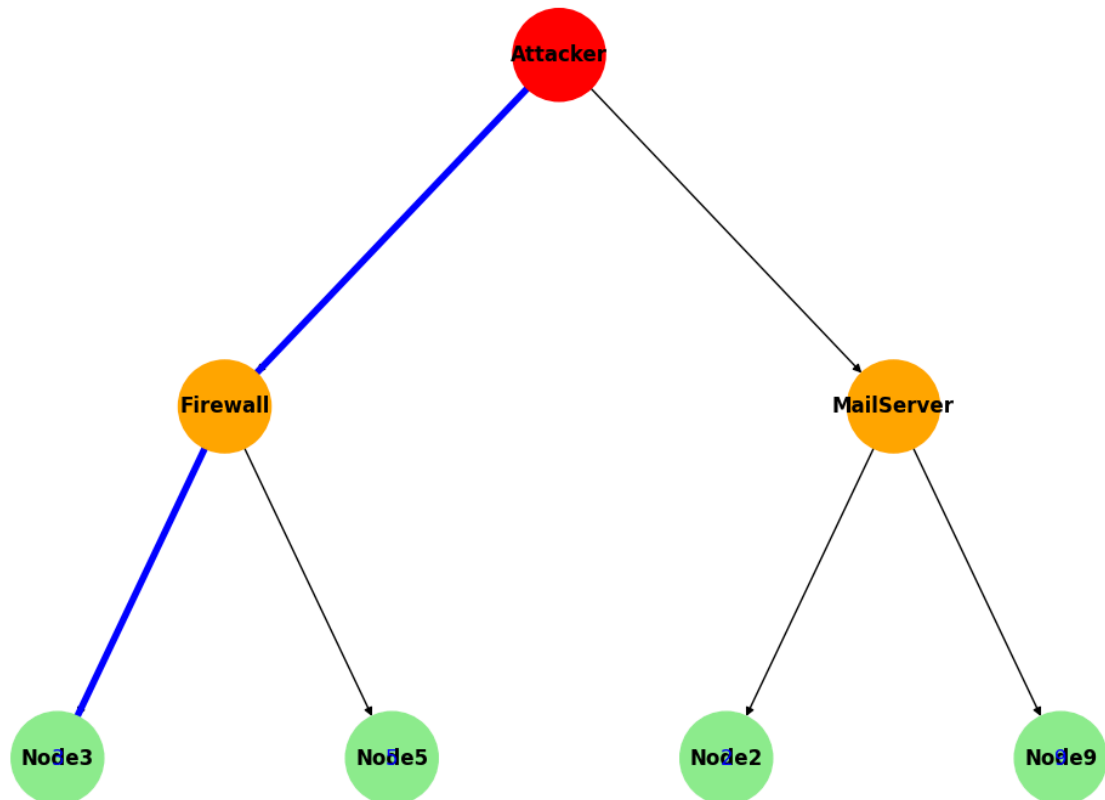
best_edges = list(zip(best_path, best_path[1:]))

nx.draw_networkx_edges(
    G,
    pos,
    edgelist=best_edges,
    edge_color="blue",
    width=4
)

plt.title("Minimax Algorithm - Optimal Path Highlighted", fontsize=14, fontweight="bold")
plt.axis("off")
plt.show()
```

```
=====
MINIMAX RESULT
=====
Optimal Value: 3
Best Path: Attacker → Firewall → Node3
```

Minimax Algorithm - Optimal Path Highlighted



✓ ALPHA-BETA PRUNING

```
import networkx as nx
import matplotlib.pyplot as plt

# -----
# GAME TREE
# -----

G = nx.DiGraph()

edges = [

    ("Attacker", "Firewall"),
    ("Attacker", "MailServer"),

    ("Firewall", "Node3"),
    ("Firewall", "Node5"),

    ("MailServer", "Node2"),
    ("MailServer", "Node9")
]
```

```

]

G.add_edges_from(edges)

values = {
    "Node3": 3,
    "Node5": 5,
    "Node2": 2,
    "Node9": 9
}

# =====
# TRACKING
# =====

visited_edges = []
pruned_edges = []

# -----
# ALPHA-BETA FUNCTION
# -----

def alphabeta(node, depth, alpha, beta, maximizingPlayer):

    if node in values:
        return values[node]

    children = list(G.successors(node))

    if maximizingPlayer:

        value = -float("inf")

        for i, child in enumerate(children):

            visited_edges.append((node, child))

            value = max(
                value,
                alphabeta(child, depth+1, alpha, beta, False)
            )

            alpha = max(alpha, value)

            # ✖ PRUNING OCCURS
            if alpha >= beta:

                for remaining in children[i+1:]:
                    pruned_edges.append((node, remaining))

                break

        return value

    else:

        value = float("inf")

        for i, child in enumerate(children):

            visited_edges.append((node, child))

            value = min(
                value,
                alphabeta(child, depth+1, alpha, beta, True)
            )

            beta = min(beta, value)

            # ✖ PRUNING OCCURS

```

```

        if beta <= alpha:

            for remaining in children[i+1:]:
                pruned_edges.append((node, remaining))

            break

        return value

# -----
# RUN
# -----

result = alphabeta(
    "Attacker",
    0,
    -float("inf"),
    float("inf"),
    True
)

print("\nRESULT:", result)
print("PRUNED EDGES:", pruned_edges)

# -----
# VISUALIZATION
# -----

plt.figure(figsize=(10,7))

pos = {

    "Attacker": (0,4),
    "Firewall": (-2,2),
    "MailServer": (2,2),
    "Node3": (-3,0),
    "Node5": (-1,0),
    "Node2": (1,0),
    "Node9": (3,0)
}

# Base graph
nx.draw(
    G,
    pos,
    with_labels=True,
    node_size=3000,
    node_color="lightblue",
    font_weight="bold"
)

# Edge labels (values)
nx.draw_networkx_edge_labels(
    G,
    pos,
    edge_labels={}
)

# -----
# VISITED PATH (BLACK)
# -----

nx.draw_networkx_edges(
    G,
    pos,
    edgelist=visited_edges,
    edge_color="black",
    width=3
)

```

```

# -----
# PRUNED PATH (RED DASHED)
# -----

nx.draw_networkx_edges(
    G,
    pos,
    edgelist=pruned_edges,
    edge_color="red",
    style="dashed",
    width=3
)

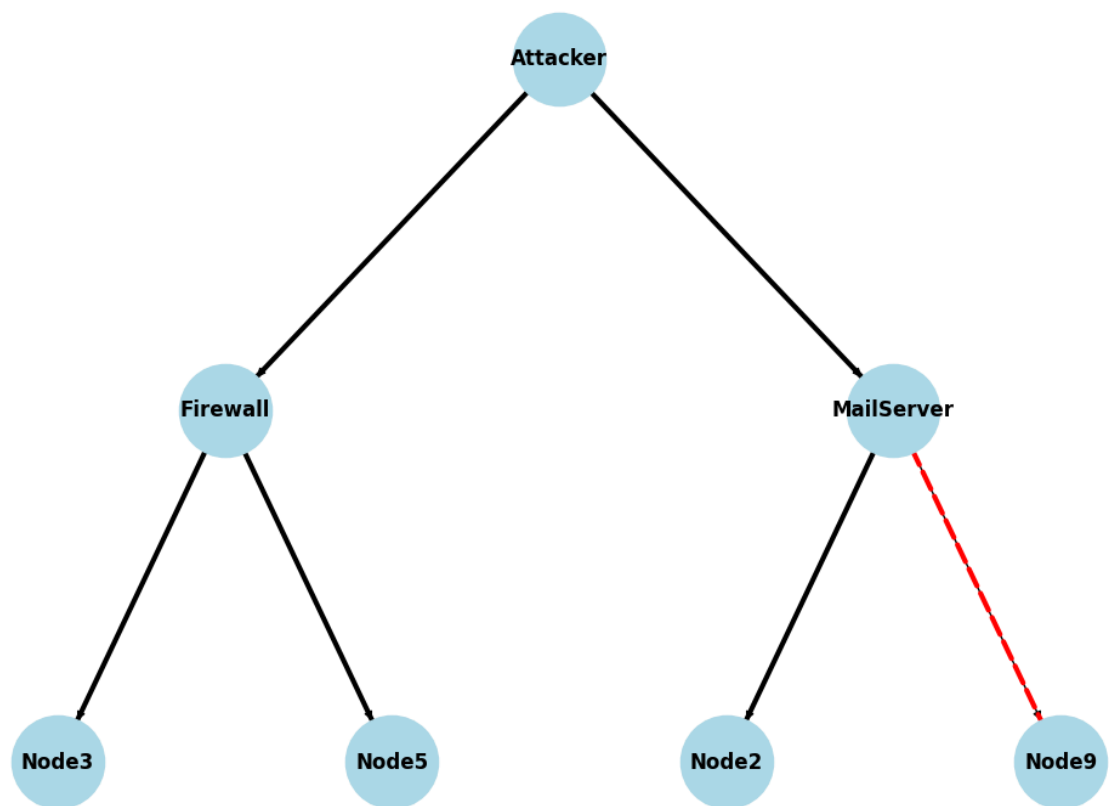
plt.title("Alpha-Beta Pruning (Correct Visualization)")
plt.axis("off")
plt.show()

```

RESULT: 3

PRUNED EDGES: [('MailServer', 'Node9')]

Alpha-Beta Pruning (Correct Visualization)



✓ Output and Empirical Analysis

```

import time
import heapq
from collections import deque
import matplotlib.pyplot as plt

```